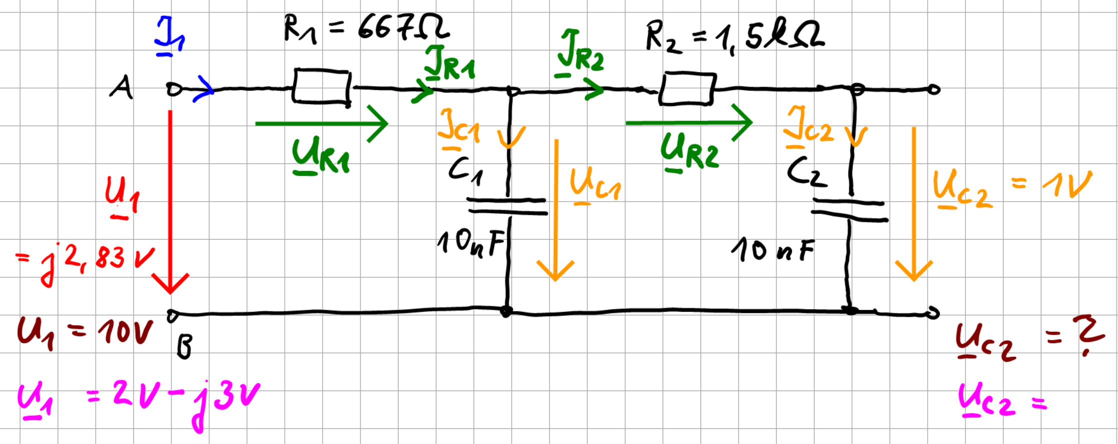
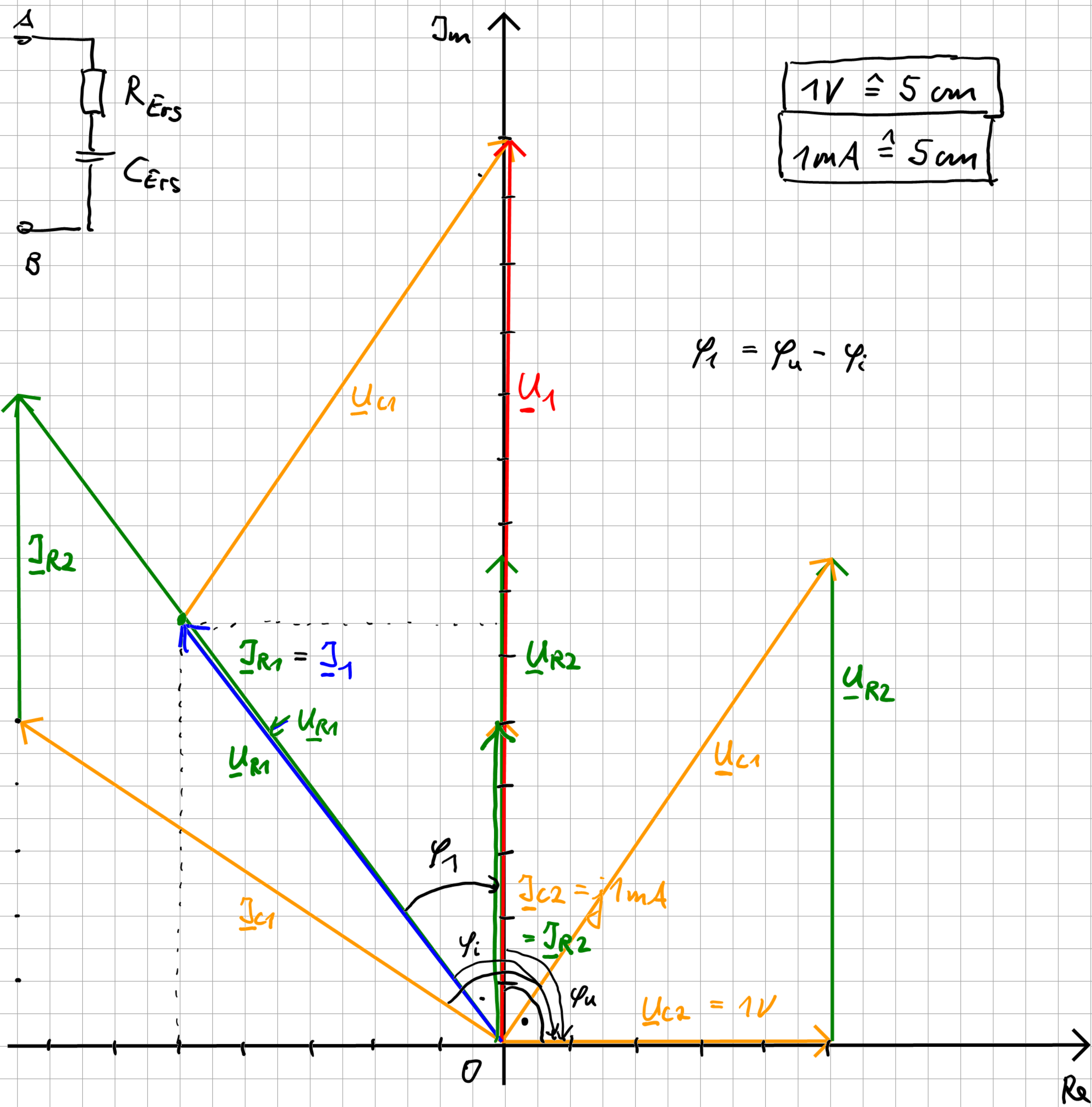
Kap. 6: Konstruktion von maßstäblichen Zeigerdiagrammen



$f = 15{,}915\ kHz$

Vorgabe: $U_{C2} \mathrel{\hat=} 5\,cm$

$I_{C2} \mathrel{\hat=} 5\,cm$



Kreisfrequenz:

$$\omega = 2\pi \cdot f = 2\pi \cdot 15{,}915\,kHz = 100000 \cdot \frac{1}{s} = 10^5 \cdot \frac{1}{s}$$

Bestimmung der Impedanzen:

$$\underline{Z}_{C1} = \frac{1}{j\omega C_1} = \frac{1}{j\, 10^5 \cdot \frac{1}{s} \cdot 10\,nF} = \frac{1}{j}\, 1000\, \frac{1}{\frac{1}{s}\,\frac{As}{V}} =$$

$$= -j\, 1\,k\Omega$$

$$\underline{Z}_{C2} = \underline{Z}_{C1} = -j\, 1\,k\Omega$$

$$\underline{I}_{C2} = \frac{\underline{U}_{C2}}{\underline{Z}_{C2}} = \frac{1V}{-j\, 1k\Omega} = +j\, 1\,mA$$ ($\hat{=}$ 5 cm) Vorgabe

$$\underline{I}_{R2} = \underline{I}_{C2}$$

$$\underline{U}_{R2} = R_2 \cdot \underline{I}_{R2}$$

$$= 1{,}5\,k\Omega \cdot j\, 1\,mA$$

$$= j\, 1{,}5\,V$$ ($\hat{=}$ 7,5 cm)

$$\underline{U}_{C1} = \underline{U}_{R2} + \underline{U}_{C2}$$

$$= j\, 1{,}5V + 1V$$

$$= 1V + j\, 1{,}5\,V$$

$$\underline{I}_{C1} = \frac{\underline{U}_{C1}}{\underline{Z}_{C1}}$$

$$= \frac{1V + j\, 1{,}5V}{-j\, 1\,k\Omega} = \frac{1V}{-j\, 1k\Omega} + \frac{j\, 1{,}5V}{-j\, 1k\Omega}$$

$$= +j\, 1\,mA - 1{,}5\,mA$$

$$\underline{I}_1 = \underline{I}_{R1} = \underline{I}_{C1} + \underline{I}_{R2}$$

$$= j\, 1\,mA - 1{,}5\,mA + j\, 1\,mA$$

$$= -1{,}5\,mA + j\,2\,mA$$

↑ Re < 0

$$= \sqrt{(-1{,}5mA)^2 + (2mA)^2} \cdot e^{j\arctan\left(\frac{2mA}{-1{,}5mA}\right) \pm j180°}$$

$$= 2{,}5\,mA \cdot e^{-j(53° \pm j180°)}$$

$$= 2{,}5\,mA \cdot e^{+j\,126{,}9°}$$

$$\underline{U}_{R1} = R_1 \cdot \underline{I}_{R1}$$

$$= 667\Omega \cdot (-1{,}5mA + j\,2mA)$$

$$= -1\,V + j\,1{,}334\,V$$

$$= -1\,V + j\,\frac{4}{3}\,V$$

$$\underline{U}_1 = \underline{U}_{R1} + \underline{U}_{C1}$$

$$= -1V + j\,\frac{4}{3}V + 1V + j\,1{,}5V$$

$$= j\,2{,}833\,V$$

$1V \mathrel{\hat{=}} 5\,cm$

$\Rightarrow \quad U_1 \mathrel{\hat{=}} 14{,}2\,cm$

Phasenverschiebung am Eingang

$$\varphi_1 = \varphi_u - \varphi_i$$

$$\underline{U}_1 = j\,2{,}833\,V \quad \Rightarrow \quad \varphi_{u1} = 90°$$

$$\underline{I}_1 = 2{,}5mA \cdot e^{j\,126°} \quad \Rightarrow \quad \varphi_{i1} = 126°$$

$$\varphi_1 = 90° - 126°$$

$$= -36°$$

Linearität

$$\underline{U}_{C2} = 1V \quad \Rightarrow \quad \underline{U}_1 = j\,2{,}83V$$

$$\times \quad \Leftarrow \quad \underline{U}_1 = 10V$$

$$\frac{\underline{U}_{C2}}{X} = \frac{j\,2{,}83V}{10V}$$

$$\Rightarrow \quad X = \frac{10V \cdot \underline{U}_{C2}}{j\,2{,}83V} = \frac{10V \cdot 1V}{j\,2{,}83V} = 3{,}53V\,\frac{1}{j} = -j\,3{,}53V$$

$\underline{U}_1 = j\,2{,}83$ $\qquad\qquad$ $\underline{U}_{C2} = 1V$

$\underline{U}_1 = 2V - j3V$ $\qquad\qquad$ $\underline{U}_{C2} = X$

$$\frac{j\,2{,}83\,V}{2V - j3V} = \frac{1V}{X}$$

$$\Rightarrow \quad X = \frac{(2V - j3V)\cdot 1V}{j\,2{,}83V}$$

$$= \frac{1}{j}\,0{,}707V - 1{,}06\,V$$

$$= -1{,}06\,V - j\,0{,}707V$$

Re < 0

$$= \sqrt{(1{,}06V)^2 + (0{,}707V)^2} \cdot e^{j\left(\arctan\left(\frac{-0{,}707}{-1{,}06}\right) \pm 180°\right)}$$

$$= 1{,}27\,V \cdot e^{-j\,146°}$$